



2026

Chemistry

Paper 2 — Covalent Bonding · Acids & Bases · Neutralisation

Time allowed: 45 minutes

Total marks: 50

Index Number: _____

Instructions to Candidates

1. Write your Index Number in the space provided above.
2. Answer **all** questions in **all three sections**.
3. Section A: Choose **one** answer for each question. Tick the box next to your chosen answer.
4. Section B & C: Write your answers in the spaces provided.
5. Use a blue or black pen. Diagrams may be drawn in pencil.
6. All working must be shown for calculations.
7. The number of marks is given in brackets [] at the end of each question.

Section	Marks Available	Marks Awarded
A	15	
B	20	
C	15	
Total	50	

SECTION A — Multiple Choice [15 marks] Answer all 15 questions. Tick ONE box per question.

- 1 In a covalent bond, the atoms are held together by [1]
- ☐ A transfer of electrons from one atom to another
- ☐ B sharing of a pair of electrons between the two atoms
- ☐ C delocalised electrons moving through a lattice
- ☐ D electrostatic attraction between oppositely charged ions
- 2 Which of the following molecules contains a **double** covalent bond? [1]
- ☐ A H_2
- ☐ B HCl
- ☐ C O_2
- ☐ D NH_3
- 3 How many covalent bonds are present in one molecule of water (H_2O)? [1]
- ☐ A 1
- ☐ B 2
- ☐ C 3
- ☐ D 4
- 4 Carbon dioxide (CO_2) has the shape [1]
- ☐ A linear ($\text{O}=\text{C}=\text{O}$ in a straight line)
- ☐ B bent / V-shaped
- ☐ C trigonal planar
- ☐ D tetrahedral
- 5 A student adds a few drops of universal indicator to an unknown solution. The indicator turns red. The solution is [1]
- ☐ A strongly acidic
- ☐ B weakly acidic
- ☐ C neutral
- ☐ D alkaline
- 6 The pH of a neutral aqueous solution at room temperature is [1]
- ☐ A 0

- ☐ B 5
☐ C 7
☐ D 14

7 Which of the following is a **strong acid**?

[1]

- ☐ A Hydrochloric acid (HCl)
☐ B Ethanoic acid (CH_3COOH)
☐ C Citric acid
☐ D Carbonic acid (H_2CO_3)

8 A base is best defined as a substance that

[1]

- ☐ A turns litmus red
☐ B neutralises an acid to form a salt and water only
☐ C has a pH below 7
☐ D dissolves in water to produce H^+ ions

9 When a metal carbonate reacts with dilute hydrochloric acid, the products are

[1]

- ☐ A salt + water
☐ B salt + hydrogen
☐ C salt + water + carbon dioxide
☐ D oxide + water

10 The pH of a sodium hydroxide (NaOH) solution is

[1]

- ☐ A exactly 7
☐ B between 1 and 6
☐ C between 7 and 14 (above 7)
☐ D always exactly 14

11 The neutralisation reaction between HCl and NaOH always produces

[1]

- ☐ A hydrogen gas and a metal oxide
☐ B a salt and water
☐ C a salt and carbon dioxide
☐ D a salt, water, and hydrogen

12 Methane (CH_4) has which type of bonding?

[1]

- ☐ A Ionic bonding only
☐ B Covalent bonding only
☐ C Metallic bonding

☐ D Both ionic and covalent bonding

13 Which of these is a **weak acid**?

[1]

- ☐ A Sulfuric acid (H_2SO_4)
- ☐ B Nitric acid (HNO_3)
- ☐ C Ethanoic acid (CH_3COOH)
- ☐ D Hydrochloric acid (HCl)

14 What are the products when hydrochloric acid reacts with sodium hydroxide solution?

[1]

- ☐ A $\text{NaCl} + \text{H}_2\text{O}$
- ☐ B $\text{NaCl} + \text{H}_2$
- ☐ C $\text{NaOH} + \text{HCl}$
- ☐ D $\text{Na}_2\text{O} + \text{H}_2\text{O}$

15 How many shared pairs of electrons (bonds) are present in a molecule of nitrogen (N_2)?

[1]

- ☐ A 1 (single bond)
- ☐ B 2 (double bond)
- ☐ C 3 (triple bond)
- ☐ D 4

SECTION B — Short Structured Questions [20 marks] Answer all questions.

B1 This question is about covalent bonding.

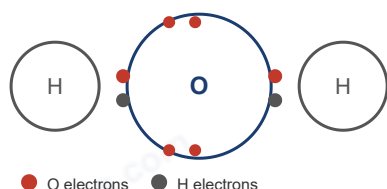


Fig. 1 — Dot-and-cross diagram of water (H_2O). Only outer-shell electrons are shown.

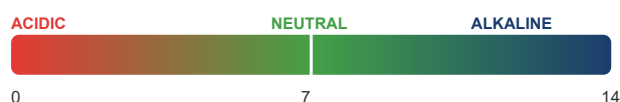
- (a) Using Fig. 1, state the number of covalent bonds in one molecule of water. [1]

- (b) In the box below, draw a dot-and-cross diagram for a molecule of ammonia (NH_3). Nitrogen has 5 electrons in its outer shell. Show outer-shell electrons only. [3]

Draw your diagram here

- (c) State one difference between covalent bonding and ionic bonding. [1]

B2 This question is about acids and bases.



- (a) A solution has pH 2. State whether this solution is acidic, neutral, or alkaline. Name **one** laboratory acid that would have a pH of approximately 2. [2]

- (b) A student has solutions of hydrochloric acid (pH 1) and ethanoic acid (pH 3), both at the same concentration. Explain why hydrochloric acid has a lower pH than ethanoic acid. [2]

- (c) Name the ion responsible for the acidic properties of aqueous acid solutions. [1]

B3 This question is about the preparation of copper sulfate (CuSO_4) crystals by neutralisation.

- (a) Name the acid and the base used to prepare copper sulfate solution by neutralisation. [2]

Acid:

Base:

- (b) Write the balanced chemical equation for the reaction between copper(II) oxide and dilute sulfuric acid. Include state symbols. [2]

- (c) Describe how you would obtain dry copper sulfate crystals from the solution produced.

[1]

B4 This question is about neutralisation reactions.

- (a) Write the balanced chemical equation for the reaction between hydrochloric acid (HCl) and sodium hydroxide (NaOH).

[1]

- (b) Write the balanced chemical equation for the reaction between calcium carbonate (CaCO_3) and excess dilute hydrochloric acid (HCl). Include state symbols.

[2]

- (c) Write the ionic equation for the neutralisation of any strong acid by any strong alkali. Include state symbols.

[2]

SECTION C — Long Structured Questions [15 marks] Answer all questions.

C1 Covalent Bonding and Properties of Simple Molecules

[8]

Carbon dioxide (CO₂) and water (H₂O) are both covalently bonded molecular compounds. Their properties are compared in the table below.

Property	CO ₂	H ₂ O
Melting point	-78 °C (sublimes)	0 °C
Boiling point	-57 °C	100 °C
Conducts electricity (liquid)	No	Very slightly
Number of bonds	2 double bonds	2 single bonds

- (a) In the box below, draw a dot-and-cross diagram for CO₂. Show outer-shell electrons only. (Carbon has 4 outer electrons; oxygen has 6.)

[2]

Draw your diagram here

- (b) Explain why both CO₂ and H₂O have low melting and boiling points compared with ionic compounds such as NaCl.

[2]

- (c) Explain why neither CO₂ nor H₂O (pure liquid) conducts electricity well.

[1]

- (d) Carbon dioxide dissolves in water to form a solution that turns litmus **red**. Write the word equation for this reaction.

[1]

- (e) State the approximate pH of the solution formed when carbon dioxide dissolves in water. Give a reason for your answer. [2]

C2 Acids, Bases, and Preparation of Salts [7]

A student prepares zinc sulfate (ZnSO_4) by reacting excess zinc oxide powder with dilute sulfuric acid.

- (a) Write the balanced chemical equation for the reaction between zinc oxide and dilute sulfuric acid. [2]

- (b) The student uses **excess** zinc oxide. Explain the purpose of using excess zinc oxide in this preparation. [1]

- (c) Describe how the student would obtain the zinc sulfate solution from the mixture after the reaction is complete. [1]

- (d) Describe how the student would then obtain dry zinc sulfate crystals from the solution. Include the names of the two techniques used. [2]

- (e) State whether zinc sulfate solution would be acidic, alkaline, or neutral. Give a reason.

[1]

Mark Scheme — Paper 2

For teacher/self-assessment use only. Accept any chemically correct equivalent.
Balanced equations: all species must be correct; one mark for formula, one for balancing unless stated otherwise. State symbols only required where the question asks for them.

Section A — Multiple Choice

Q	Answer	Q	Answer	Q	Answer
1	B	6	C	11	B
2	C	7	A	12	B
3	B	8	B	13	C
4	A	9	C	14	A
5	A	10	C	15	C

Section B

Question	Mark-worthy points	Marks
B1(a)	2	1
B1(b)	N at centre with 3 shared pairs (each a dot + cross) connecting 3 H atoms [2]; one lone pair on N [1]	3
B1(c)	In covalent bonding electrons are <u>shared</u> ; in ionic bonding electrons are <u>transferred</u> (accept: ionic compounds are made of ions / covalent are made of molecules)	1
B2(a)	Acidic [1]; hydrochloric acid / sulfuric acid / nitric acid (any one strong acid) [1]	2
B2(b)	HCl is a <u>strong</u> acid — fully dissociates into H ⁺ ions [1]; ethanoic acid is a <u>weak</u> acid — only partially dissociates, so fewer H ⁺ ions, higher pH [1]	2
B2(c)	H ⁺ (hydrogen / oxonium ions; accept H ₃ O ⁺)	1
B3(a)	Acid: sulfuric acid (H ₂ SO ₄) [1]; Base: copper(II) oxide (CuO) [1]	2
B3(b)	CuO(s) + H ₂ SO ₄ (aq) → CuSO ₄ (aq) + H ₂ O(l) [1 for correct formulae, 1 for balanced with state symbols]	2
B3(c)	Evaporate the solution (by heating) to concentrate it / until crystals appear; leave to crystallise; filter/dry the crystals	1
B4(a)	HCl + NaOH → NaCl + H ₂ O (or HCl(aq) + NaOH(aq) → NaCl(aq) + H ₂ O(l))	1
B4(b)	CaCO ₃ (s) + 2HCl(aq) → CaCl ₂ (aq) + H ₂ O(l) + CO ₂ (g) [1 for formulae, 1 for balancing + state symbols]	2

Question	Mark-worthy points	Marks
B4(c)	$\text{H}^+(\text{aq}) + \text{OH}^-(\text{aq}) \rightarrow \text{H}_2\text{O}(\text{l})$ [1 for correct ions, 1 for state symbols]	2

Section C

Question	Mark-worthy points	Marks
C1(a)	C at centre [1]; each O connected to C by 2 shared pairs (double bond) [1]; correct lone pairs on each O [bonus — not required for full marks]	2
C1(b)	Simple/molecular covalent compounds — only <u>weak intermolecular forces</u> between molecules [1]; little energy needed to overcome these forces, so low melting/boiling points [1] (contrast: ionic compounds have strong electrostatic bonds in giant lattice)	2
C1(c)	No free ions / no charged particles free to move; covalent molecules are not charged	1
C1(d)	Carbon dioxide + water $\rightarrow$ carbonic acid	1
C1(e)	pH approximately 5–6 (accept 4–6) [1]; because carbonic acid is a <u>weak acid</u> — partially dissociates, producing H^+ ions [1]	2
C2(a)	$\text{ZnO} + \text{H}_2\text{SO}_4 \rightarrow \text{ZnSO}_4 + \text{H}_2\text{O}$ [1 for correct formulae, 1 for balanced]	2
C2(b)	To ensure all the acid is used up / neutralised; so the product is not contaminated with excess acid	1
C2(c)	Filter the mixture to remove excess unreacted zinc oxide	1
C2(d)	Evaporation (heat solution to concentrate/reduce volume) [1]; crystallisation (allow to cool so crystals form, then filter/dry) [1]	2
C2(e)	Neutral (accept slightly acidic for 1 mark with valid reason) [1]; zinc sulfate is a salt formed from a strong acid and a base — it does not hydrolyse significantly at NCE level [1]	1